Sophia S. Liau

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Interested in software development, data engineering, and biomedical technology.

Education

University of British Columbia Sept 2023 – Present
BASc in Biomedical Engineering; Specialization in Bioinformatics, Minor in Commerce
Relevant Coursework: Biomedical Engineering Design, Software Construction, Algorithms & Data Structures

Experience

Engineering Co-op - Asset Management, UBC Municipal – Vancouver, BC May 2026 – Dec 2026

- Built a **Python pipeline** to organize raw contractor storm drain inspection files, catch errors and store data in ArcGIS via a **REST API**.
- Developed **88 automated tests** for the pipeline, including GIS integration, validation, and database workflows.
- Presented a dashboard and a state-of-infrastructure report, visualizing inspection and risk status across **3,289 assets** to support capital planning.
- Designed a **database on Microsoft SQL Server for 50,000+ assets** to streamline data governance.
- Engaged with stakeholders to audit current safety habits and sentiments, informing project direction and a **40+ participant safe-work system pilot** ahead of an organization-wide rollout next year.
- Developed a standardized safety system across six operational groups, coordinating with frontline teams.
- Drafted CAD drawings in accordance with accessible building guidelines and provincial municipal regulations.

Events and Outreach Coordinator, Railtown AI – Vancouver, BC May 2026 – Present

- Initiated outreach with various external groups, including student clubs and software communities in Vancouver.
- Executed 2 major agentic AI events with **100+ participants,** organizing marketing, content themes and catering.
- Created strategies to grow our user base, resulting in a **total of 234 GitHub stars** on our open source framework.

Coding Instructor, UTG Academy – Vancouver, BC Nov 2025 – Feb 2026

- Taught Python programming and foundational **AI concepts to 60+ students** ages 10–12 through project-based learning, supporting a British Columbia government initiative to improve youth technology literacy.

Projects

Liver Hepatocellular Carcinoma Analysis – BMEG 310 Oct 2025 - Dec 2025

- Conducted a study on mutation, clinical and sequencing patterns in hepatocellular carcinoma using TCGA data.
- Wrote an R analysis pipeline to connect genomic variation and gene-expression structure with patient characteristics, cancer stage and multiple survival outcomes via **PCA, clustering and survival analysis**.

Personal Website – Personal Project Dec 2024 – Present

- Designed a personal portfolio website using Figma, following UI/UX principles while conveying personality.
- Built a stylized website with **HTML/CSS and JavaScript**, and deployed it on Cloudflare.

Leadership

VP Academic, Biomedical Engineering Undergraduate Student Association May 2026 – Present

- Represented undergraduate biomedical engineering students on the Academic Committee and communicating student priorities, course concerns, and recommendations directly to faculty and School leadership.

VP Social, Biomedical Engineering Undergraduate Student Association May 2025 – Oct 2026

- Planned and delivered 12+ student events, including interdepartmental parties and a formal, managing end-to-end logistics, sponsorship outreach, budgeting, facility approvals, marketing, and execution.

Skills

Programming Languages: Python, Java, JavaScript, MATLAB, HTML/CSS, C, C++, R, SQL
Domains: Bioinformatics, Data Engineering, GIS & Spatial Analytics, Asset Management
Tools: Git, MS Office, ArcGIS, Figma, AutoCAD, SolidWorks, SQL Server, Jupyter Notebook, VS Code, KiCad
Languages Spoken: English, French, Japanese